

Smart Lender

AI-Based Loan Eligibility Prediction System Documentation

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Track: AI-ML and Gen AI Track

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1. System Overview

The **Raising Water Monitoring & Flood Alert System** is an AI-powered web application designed to monitor real-time water levels and predict flood risks using IoT sensors, weather data, and Machine Learning. The system integrates a Flask-based web application with a MySQL database to collect, process, and analyze water level readings, rainfall information, and historical flood data.

The application continuously monitors water levels and predicts the probability of flooding before it occurs. Users receive instant alerts through the dashboard and notification system, enabling authorities and residents to take preventive measures in advance.

The system minimizes manual monitoring, improves prediction accuracy, enhances disaster preparedness, and helps protect lives and property.

2. System Architecture

The application follows a standard layered architecture:

- **Presentation Layer** – HTML5, CSS3, JavaScript, Bootstrap
- **Application Layer** – Flask REST API
- **Business Logic Layer** – Authentication, Sensor Validation, Alert management
- **Machine Learning Layer** – Scikit-learn Flood Prediction Model
- **IoT Layer** – ESP32 / NodeMCU, Ultrasonic Water Level Sensor
- **Weather Integration Layer** – OpenWeather API
- **Data Layer** – MySQL Database with SQLAlchemy ORM

POST /api/predict

Receives water level and weather information, validates sensor data, predicts flood risk using the trained Machine Learning model, stores the prediction, and returns the flood risk level.

Request Format:-

```
{  
  "location": "Kurnool",  
  "water_level": 8.4,  
  "rainfall": 62,  
  "temperature": 29,
```

```
"humidity":82,  
"river_flow":165,  
"wind_speed":14  
}
```

Response Format:-

```
{  
  "risk_level":"High",  
  "flood_probability":"93.4%",  
  "alert":"Flood Warning",  
  "recommendation":"Evacuate Low-Lying Areas"  
}
```

3. Setup & Installation Instructions

Follow these steps to set up and run the Vanguard HDI application locally:

- **Step 1 – Create Virtual Environment:-**
python -m venv venv
- **Step 2 - Activate Virtual Environment:-**
venv\Scripts\activate
- **Step 3 – Install Dependencies:-** pip install -r requirements.txt
- **Step 4 – Configure Database:-** Create a MySQL database named: smart_lender
Update database credentials inside config.py
- **Step 5 – Train the Machine Learning Model:-** python Training/train_model.py
loan_model.pkl
- **Step 6 – Run the Flask Application:-** python app.py
- **Step 7 – Open the Application:-**
http://127.0.0.1:5000/